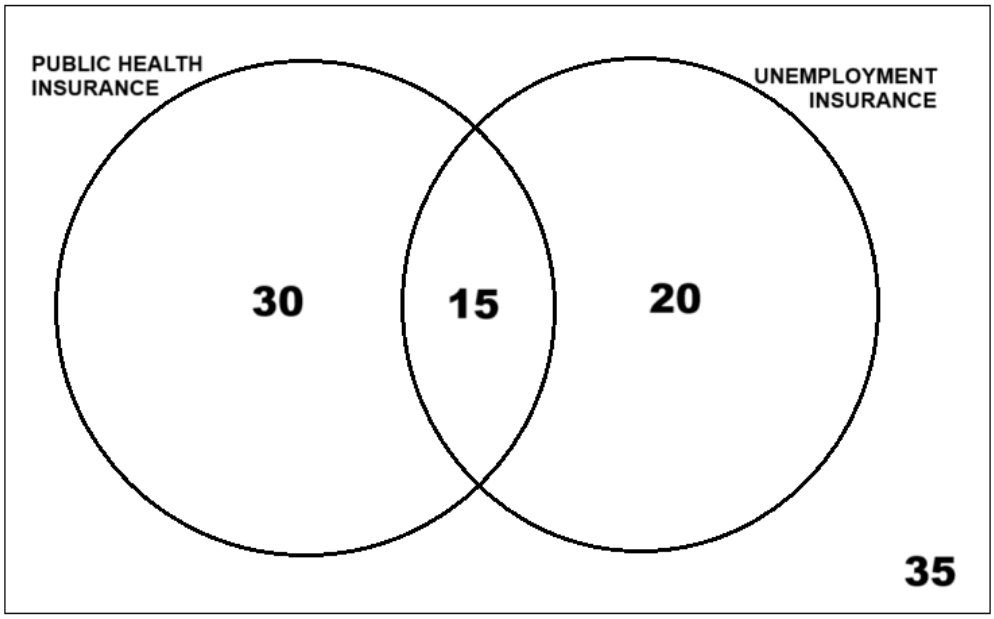
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EXERCISE SHEET / THEME 2: MULTIPLE CATEGORICAL VARIABLES.

EXERCISE 1. VENN DIAGRAM AND INSURANCE COVERAGE.

Context: the following Venn Diagram presents the distribution of a group of surveyed people. One group represents those covered by a public health insurance (PHI), while the other represents those who are part of the unemployment insurance (UI).



Considering the diagram, answer the following questions:

a) Find the cardinality (i.e. the number of elements) of the set representing individuals who have at least one form of insurance: (PHI ∪ UI)?
b) Find the cardinality of the set representing individuals who have both forms of insurance: (PHI ∩ UI)?
c) Find the cardinality of the set representing individuals who have unemployment insurance: (UI)?
d) Find the cardinality of the set representing individuals who don't have unemployment insurance: (UI')?
e) Find the cardinality of the set representing individuals who do not have any form of insurance: (PHI ∪ UI)'?
f) Find the cardinality of the set representing individuals with only one insurance: (PHI ∩ GHI)'?
g) Find the total number of individuals surveyed in the society: (U)?

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EXERCISE 2. CONTINGENCY TABLE AND CARE DUTIES.

Table. Frequency table showing the care duties by gender.

Gender	Performs	Frequency
Female	Yes	35
Female	No	5
Male	Yes	25
Male	No	75

a) Fill a contingency table like the following with absolute frequencies.

		Perform care duties	
		Yes	**No**
Gender	**Female**		
	Male		

b) Calculate the proportions **by gender** and fill a conditional percentage distribution table like the following. Consider that each gender represents its own 100%.

		Perform care duties		Percentage
		Yes	**No**	
Gender	**Female**			100
	Male			100

c) Create a mosaic plot based on your first answer (Part A. Contingency Table with Absolute Frequencies).

d) Compare the information that can be extracted from
   i. The mosaic plot that you just plotted, and
   ii. The following cumulated bar plot given (which is based on the information that you should have in the Part B. Table with Conditional Distributions).

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